

Towards a Learning Analytics Support for Intelligent Tutoring Systems on MOOC Platforms

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Abstract—Nowadays, many MOOC platforms have arisen to provide free knowledge. These platforms have a large catalog of courses for different specializations that progressively demand more specific learning resources and assessment methods to evaluate the progression of students. Current MOOC platforms are gradually giving support to these new requirements but with a limited assistance. This paper presents the state of art of the analytical system for three relevant MOOC platforms, one of the main pillars for analyzing the progression of courses. Other initiatives are also reviewed to show that current MOOC analytical systems are not ready to support custom MOOC-aware intelligent tutoring systems (ITSs). Thus, the design of a learning analytics system to assist these tools for MOOC platforms is presented.

Keywords—Learning analytics, MOOC, intelligent tutoring systems, Coursera, edX, Canvas.

I. INTRODUCTION

MOOC (or Massive Online Open Courses) is an evolution of distance learning to provide free knowledge. During the past years, a considerable number of MOOC platforms have appeared, such as Coursera, edX, Khan Academy, Udacity or Canvas, and, consequently, there was an explosion of free courses. These courses have hundreds of students learning from online resources mostly in video format and they are assessed with automatic tools mostly in quiz format or peer review assessment.

Similar to any online course in a LMS, instructors need real time information about the learning process on the course becoming a challenging problem due to the process of large amount of data. MOOC platforms handle hundreds of courses with many learners (e.g. Coursera had more than 12 million unique registered learners and more than 1000 courses on 2015). Therefore, the data warehouse should be ready to handle and process all the stored information.

Learning analytics support on these platforms is being improved on each new release. However, the current analytical reports are far from being a tool to completely analyze all faces of a MOOC course (learning resources, assessment, communication, correlation among indicators, etc). Thus, we can find in the literature many innovation actions to provide specific case studies on MOOCs. Some examples can be found in [1][2][3][4][5][6][7][8] which share a common

conclusion: *the support to learning analytics in MOOC platform is limited*. These contributions provide particular examples of analytical research by processing and analyzing data exported from the data warehouse of MOOC platforms. All these approaches are disconnected. But they solve common questions arisen by particular instructors' needs. Forum interaction, learning resources utilization, video playback, individual learner progression and retention are the most common concerns of instructors.

Additionally, some courses started to use custom learning resources, such as intelligent tutoring systems (ITS) to improve the learner experience in the course. For instance, an automated essay scoring is proposed in [9] or CTAT [10] is proposed to support the creation of flexible tutors for both simple and complex problem solving (mathematics, thermodynamics, programming, etc). Other courses have automated assessment systems with personalized feedback, such as *Introduction to Spark*¹ on edX to learn big data processing, *Engineering as a Service*² on edX to learn Ruby on Rails, or *Data Science Specialization*³ on Coursera to learn R programming.

Therefore, learning analytics continues ongoing trends on interactive learning analysis to apply them both in traditional ITS and in the new MOOCs, where interaction data is generated at a very large scale ("big data"). In this sense, learning analytics is built upon previous works [11], in which the interactive processing of learning data has been analyzed exhaustively [12]. There are many results and applications in education in general and MOOCs in particular, such as providing effective assessment and automatic feedback to students; supporting instructors by monitoring and predicting student performance; and modeling user profiles for content and process customization, according to the student needs, goals and individual skills [13].

However, among other issues, there is a lack of integral solutions that can support all the mentioned aspects and incorporate them openly and transparently in current ITS and MOOC platforms. Moreover, there is a lack of a common data model for representing student interactions (typically each MOOC platform uses its own

¹ <https://www.edx.org/course/introduction-spark-uc-berkeleyx-cs105x>

² <https://www.edx.org/course/engineering-software-service-saas-part-1-uc-berkeleyx-cs169-1x>

³ <https://www.coursera.org/specializations/jhu-data-science>

TABLE I. MOOC PROVIDERS LEARNING ANALYTICS COMPARISON BASED ON DYCKHOFF'S CLASSIFICATION [14].

Category	Indicator	Coursera	edX Insights	Canvas
Qualitative	Contents opinion of the course	NO	NO	NO
Quantitative	Resources utilization by day	NO	NO	NO
	Resources utilization by week	YES	YES	YES
	Resources not used	INDIRECT	INDIRECT	INDIRECT
Student comparison	Which groups can the students be classified in?	YES (active, view lecture, submit activity, forum participation)	YES (active, view lecture, submit activity)	NO
	Native vs non-native speakers	NO	NO	NO
	Student resources utilization classified by learner profile information (such as, prior learning)	NO	NO	NO
Learning resources	Resource utilization compared by type	INDIRECT (lecture, peer assessment, quiz)	INDIRECT (video playback, activity)	YES
	Relationship between learning resources and assessment activities	YES	YES	NO
Student performance	Resources used by student	NO	NO	NO
	Pass rate for assessment activity	YES	YES	YES
	Relationship among learning resources, assessment activities and forum participation	NO	NO	NO
Retention	Progression through the course	YES	YES	YES

model) which hinders the construction of a learning analytics model that manages information from different sources.

The paper is organized as follows. Section II introduces the current state of the learning analytics module on MOOC platforms where Coursera, edX and Canvas are compared. Section III reviews the literature related to learning analytics systems for MOOCs. The design of a learning analytics system to co-analyze instructional process and ITS systems for MOOCs is described in Section IV. Finally, conclusions and future work are summarized in Section V.

II. LEARNING ANALYTICS SUPPORT ON MOOC PLATFORMS

This section describes current learning analytics support on relevant MOOC platforms: Coursera and edX. We added Canvas Network since the increment of its relevance in 2015. The comparison could be extensively done by others such as Khan Academy or Udacity. However, these three selected platforms give sufficient insights of the current state of learning analytics systems on MOOC providers.

An analysis is performed based on the indicators defined by Dyckhoff [14]. The different indicators were extracted from common research questions instructors analyze on technology-enhanced learning (TEL) scenarios. This classification can be used to analyze learning analytics potential of current MOOC platforms.

The indicators are distributed on six evaluation categories. Note that, some questions/indicators may fit into more than one category:

- *Qualitative* aims to investigate opinion of learners by means of qualitative and quantitative surveys.
- *Quantitative* evaluates quantifiable data stored in the data warehouse or log files.
- *Student comparison* seeks insights about divergence on students by type.
- *Learning resources* analyzes the impact of the learning resources on the student performance.
- *Student performance* collects questions related exclusively to student performance by combining data from different sources and providing correlations, proportions and comparisons.
- *Retention* refers to indicators that may help to analyze drop-out issues.

The analysis of the three platforms is summarized in Table I. Note that, the edX Insights extension has been analyzed for edX platform instead of the edX current learning analytic system since the legacy system only supports the extraction of data in CSV format. Significant conclusions can be extracted from the table.

- Current visualization tools tend to only support aggregate analysis. Individual student progression is not supported except for Canvas platform. The



Figure 1. Sentiment analysis report on ALAS-KA [15]

main reason could be due to the privacy policy on edX and Coursera related to sensible data of students. However, Canvas is providing detailed information attached to the identity profile of the student since identity of the student is public for the instructor by default.

- The information is basically focused on student performance. The analytical information tends to describe the global progression of the course by different filters, such as time and assessment activities. edX gives additional information related to the responses of the students. This detailed information not given by other platforms shows the variety of correct and incorrect responses submitted by the learners.
- Learning resources analysis is the worst factor. Only global utilization by time is commonly summarized avoiding detailed categorization or comparison. Moreover, some information related to resources not used or utilization by type can be obtained indirectly by combining different reports. edX has a special report on video playback. This report shows total playback time for each video showing the number of students who do not finished the video. A second report is also available showing the total playback time for each playback second. Sections of the video with more seconds of reproduction give insights about difficulties on the understanding of the video or most relevant parts of the videos.
- Discussion forums utilization is also not supported. Interaction among students and topic discussion is not analyzed by any platform. Only some aggregate information related to total number of posts is displayed.

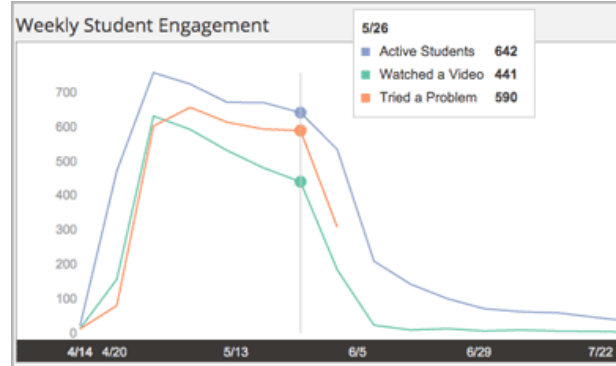


Figure 2. Student engagement report on edX Insights [18]

- Qualitative information is not currently supported by any platform. Therefore, most of the courses use external survey tools to question students at the end of the course. Currently, Coursera and edX performs a prior learning survey to give aggregate information related to the educational background of enrolled students in the course. Canvas is not supporting this feature.

Although this is the information that is currently displayed on MOOC platforms, internal data warehouses logs many of the events produced by learners (learning resources utilization, forum communication, video playback, submissions, scores, etc) and all this information can be exported. Each platform has its own process to extract this information but basically it has to be extracted by a qualified technical administrator of the platform. This requirement is basically a consequence of the privacy policy which limits the access to the sensible information of the courses.

These exports are used for external analytical tools to give more detailed information related to the instructional process. The next section performs a review on most recent external learning analytics systems for MOOC courses.

III. MOOC-BASED LEARNING ANALYTICS SYSTEMS

This section reviews the previous works related to custom learning analytics support on MOOC platforms. Aforementioned, current MOOC platforms provide little support on learning analytics information. There is a minimal interface where instructors can review some aggregated information but more fine-tuned information is not available.

The solution provided for these MOOC platforms is to make available an export feature. Log files of all the activity can be generated for external processing. Based on this information, many specific approaches have been designed to analyze the different key elements of a MOOC.



Figure 3. Temporal tag cloud visualization in [22]

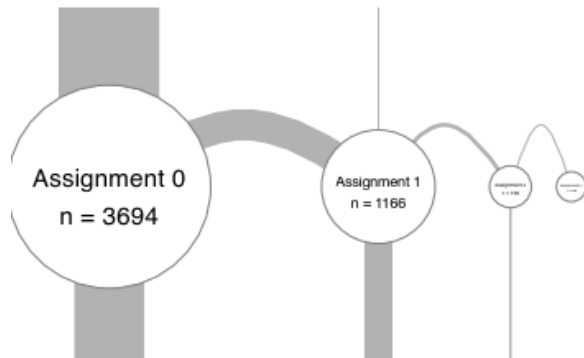


Figure 4. State transition graph report for assessment activities [23]

Some projects have been particularly designed for specific MOOC platforms. Each platform has its own data mart and these extensions are built based on these models. ALAS-KA [15] is an extension for Khan Academy platform. Although Khan Academy has a powerful learning analytics module with different visualization for learners and instructors, they are basically a direct representation of the events stored in the data mart but without any further processing. This extension processes this data to infer more valuable information related to the utilization of the platform, learning progress, timeline distribution, gamification habits, exercise solving and sentiment analysis. For example, Fig. 1 shows a sentiment analysis report based on the mood of the learner. This report is combined with relevant events in the course such as new activities, messages from instructors or publication of assessment activities scores to give insights on how these events affect to the learner.

Similarly, there are several initiatives for the edX platform. The interest for this platform is due to the existence of an open source version OpenEdx [16] which can be freely used as individual private MOOC platform. Ongoing projects, such as edX Analytics Service [17], edX Insights [18] or edX Analytics API Client [19], are currently developed and they are public available on GitHub repository. edX Insights is one the most relevant project based on processing the log files on the platform and, therefore, Section II compared this project with the current analytics modules on other platforms. The system works independently of the OpenEdx platform using an ETL (extract-transform-load) pipeline and an intermediate database to process and aggregate information. Figure 2 shows a report of student engagement. edX Analytics API Client is another project working on the edX team aiming at retrieving data directly from the edX data warehouse. Authors in [20] proposed another extension providing information about progress and timeline distribution per assignment or student. In [21], moocRP platform is

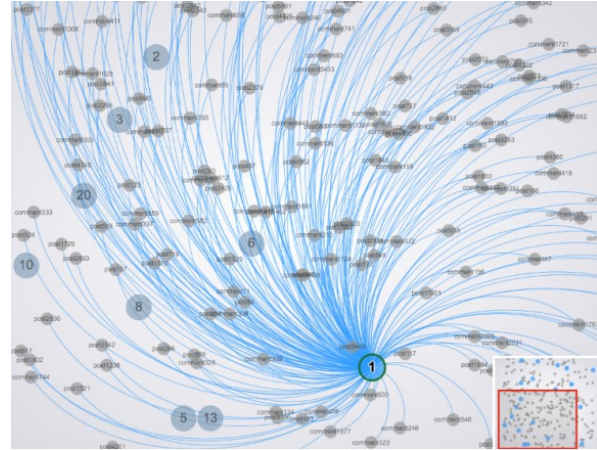


Figure 5. Discussion posts relationship report by topic [25]

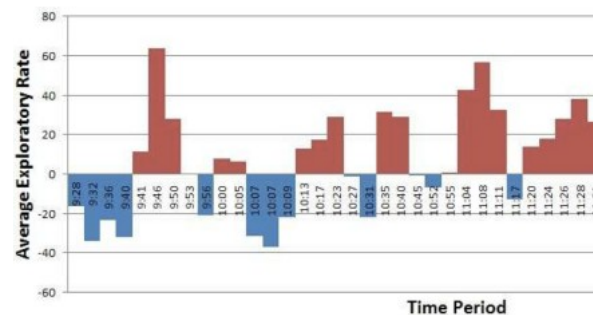


Figure 6. Timeline report of exploratory dialogue [26] proposed. Privacy, big data, surveillance and ethical issues are the main concerns of this open-source platform. Bayesian network prior-knowledge analysis and course structure content relationship are the most distinctive visualizations compared with other platforms.

For Moodle, some efforts have been also performed. For instance, a temporal tag cloud extension can be found in [22]. Figure 3 illustrates an example. This extension promotes the analysis of the most important concepts discussed in the course in a common tag cloud. However, each tag includes an individual timeline to show its progression over the time.

Coursera has also different initiatives to improve visualizations. Extensions for the platform could not be directly developed since Coursera is a private platform with no open community version. In this case, different partner institutions have developed their own analytical tools to analyze their courses. Authors in [23] designed an analytical tool for the University of Melbourne. The system offered more detailed information about student retention. Plots based on state transition graphs were used as a timeline visualization to analyze student transition from different learning resources of the course and among assessment assignments (see Fig. 4). This visualization aims to unify in one plot the student progression through the course. Similarly, University of Maryland designed an analytical system [24] to analyze correlation between used learning resources, forum participation, performed quizzes and grades obtained in the assessment activities.

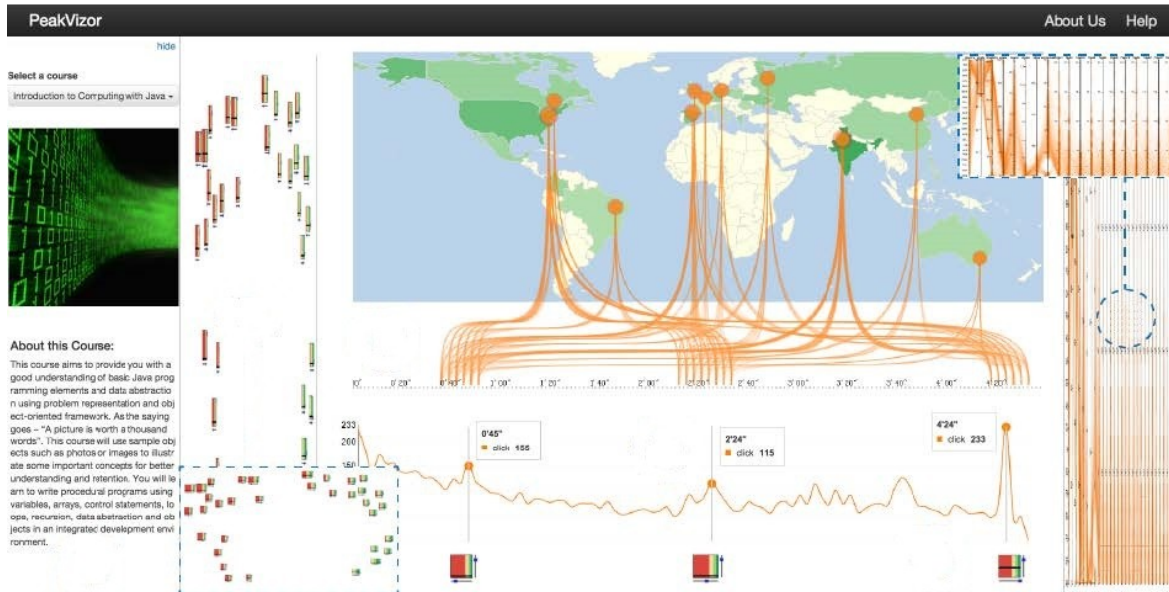


Figure 7. Peakvizor dashboard for individual video resource [31]

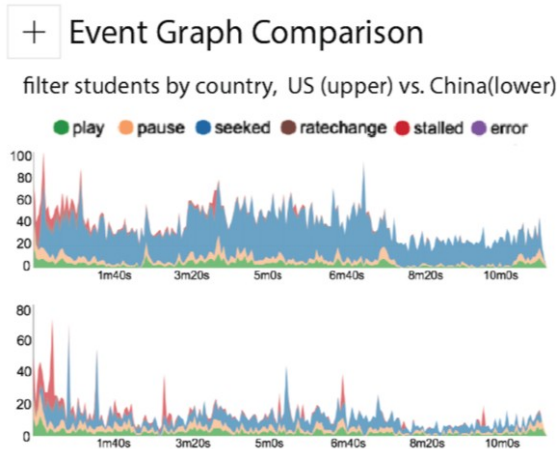


Figure 8. Individual event-graph visualization in VisMOOC [32]

These correlations help to conclude how learning resources and participation impact on the performance of students.

Related to discussion forum utilization, authors in [25] developed a set of visualization tools to analyze the forum interaction on Coursera. Topic analysis (See Fig. 5), sentiment analysis, geo-location, heat-map, word-graph are the main features. This set of visualization tools may help to analyze the interaction among learners and the topics discussed within the forum.

However, all these visualization dashboards are basically produced by processing the log files of the platform. Other authors focused on analyzing the content of the forum. Using natural language processing (NLP) techniques, authors in [26] explored the dialogue produced on the forum. Unnecessary dialogue for learning such as discussion is discarded and the technique is able to analyze the exploratory dialogue commonly associated with the construction of knowledge. Finally, the output is

summarized in different charts such as timeline progression or individual contributions to the creation of knowledge as we can observe in Fig 6. Sentiment analysis is analyzed in [27] to classify learners based on the emotional state. This information may serve as a satisfaction indicator. A different analysis is performed in [28] to predict student retention. This analysis using other models also based in NLP techniques may predict whether the learner will finish the course based on the comments posted in the forum. Another analysis and visualization tools proposed in [29] focused on classifying the posts depending on objective (question, reflection, reference, statement, feedback, etc).

Video analysis is also an important key factor on MOOCs. Most of the learning resources are based on video content and instructors needs to analyze the impact of these resources on the learning progress. Authors in [30] performed an empirical study on how video production decisions affect student engagement. Although the study did not generate any analytical tool, the approach could be adapted for creating real-time visualization charts by adding metadata information on videos description.

Video playing and interaction of the learner with the resources has been proposed in different works in the literature. Most recent approaches deal with peaks of clickstreams. PeakVizor (see Fig. 7) is proposed in [31] focusing on the utilization of glyph design to easily provide overview information of all peaks produced on a resource. Overview chart, timeline distribution and correlation between learners and peaks are the main features of the tool. In contrast, VisMOOC [32] focuses on providing more aggregated information related to events of the learner. Histogram, geo-location, timeline view of popularity or individual/global event-graph on

TABLE II. CLASSIFICATION BASED ON TEMPORAL INTERVAL INFORMATION

System/Tool	Real-time	Ending Course	Different Semesters
ALAS-KA [15]	X	X	
edX Insights [18]	X	X	
edX extension [20]	X	X	
moocRP [21]	(X)	X	
Moodle extension [22]	X	X	
MelbourneX [23]	(X)	X	
MarylandX [24]	(X)	X	
Coursera forum analysis [25]	(X)	X	
Coursera forum analysis NLP [26]	(X)	X	
Coursera Sentiment analysis [27]	(X)	X	
Coursera Student retention prediction [28]	(X)	X	
Coursera forum analysis [29]	(X)	X	
PeakVizor [31]	(X)	X	
VisMOOC [32]	(X)	X	

video playback (see Fig. 8) are the main contributions of the tool.

A. Temporal analysis

Aforementioned, standard MOOC analytics tools commonly report analytical information that is processed after the ending the course. In this section, we classify the previous systems based on the type of information that they report based on temporal intervals.

Table II reports the classification. We assume three different intervals: real-time, ending course, different sessions of the same course.

As we can see, all systems are able to provide real-time information and aggregated information at the end of the course. Real-time information has been divided in two types: system currently provides this feature (with an X) and it may provide this information (X with brackets). Note that, this availability depends on the MOOC provider export feature that is currently only available upon manual request in Coursera. Other platforms have this information available by direct connection to the data warehouse (edX, Khan) or by plugin connection (Moodle).

However, a new issue arises since comparison among semesters is not currently supported by any system. This comparison should be done explicitly by exporting different reports and comparing manually outside the systems.

B. New challenge: MOOC-based ITS learning tools

All these contributions only take into account information provided within the MOOC platforms. When a MOOC has custom learning resources or assessment

tools, platforms are not able to log their activity. The learner activity and interaction with these additional resources may be significant on practical courses since the acquisition of the practical skills or the assessment process may mainly depend on these resources. Therefore, all learning information and learners' interaction with these resources should be also collected and analyzed similar to the other generic resources of the course. As far as we know, there is no analytical tool which takes into account data generated from additional tools, such as ITS systems or external tools. Some platforms, such as Coursera, allow to use them with standard learning protocols such as LTI [33], but the activity of the external tool is not collected. This is a critical concern on current and future MOOC courses which intend to use new ways of learning and assessment not covered by the generic learning resources provided on current MOOC platforms.

Thus, analytical systems connecting ITS data utilization and MOOC platform data warehouse are required to fill this gap. The next section describes the design of a new analytical system for MOOC-aware ITS.

IV. DESIGN OF A LEARNING SUPPORT SYSTEM

In this section, we describe a new eLearning platform named ICT-FLAG [34], currently under development, aiming to provide ITSs and eLearning tools, such as MOOC platforms with innovative services in terms of learning analytics and gamification. The ICT-FLAG platform aims to support ITS that can benefit students, instructors, managers and academic coordinators, mainly through three innovative pedagogical and technological axes:

- Formative assessment tools that can provide immediate feedback by means of automatic assessment.
- Learning analytics that monitor the activity and progress of the on-line teaching and learning processes supported by eLearning tools and applications, combining this information with other sources of academic and historical information.
- Gamification as an incentive scheme in order to motivate students to practice more frequently and increase their engagement in the learning experience.

In addition to dealing with multiple components and services, the ICT-FLAG platform can support a variety of virtual campus platforms and a wide range of ITS and eLearning tools, such as Moodle.

From the architectural view, the ICT-FLAG platform provides a modular architecture with the following features:

- Customization: simple selection of which components are deployed on each course.

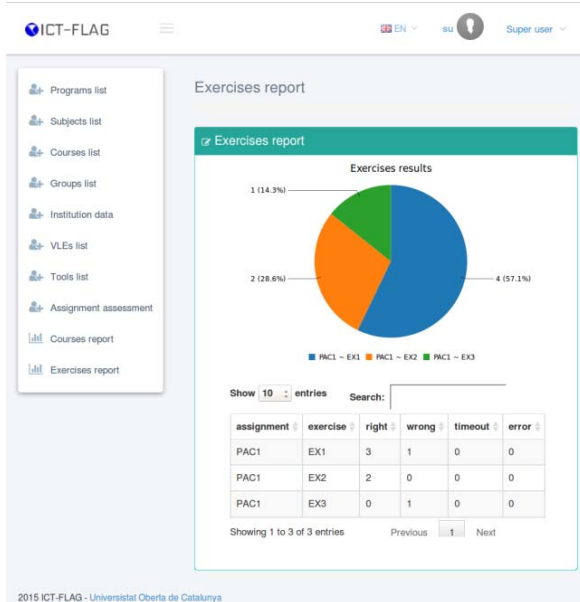


Figure 9. A report embedded into the ICT-FLAG platform.

- Distribution: to enable the distribution of components along different courses.
- Extensibility: to enable the easy extension of a course with new components.

For the sake of this paper, we describe here in more detail the Learning Analytics (LA) component of the ICT-FLAG platform. The LA component aims to process and store all the data that external tools send to the platform about learning activities and assignments (or collections of activities) in order to extract new knowledge from it (e.g., from the start date and end date of an activity we can compute its duration). The input data is stored into specific entities of the LA component and is combined with other entities of the system, such as the student who realizes the action, the assessment tool, the type of action, the activity, subject and program, and the activity's result. From this data the LA component generates reports and present information in dashboards about the eLearning progress.

The integration with current MOOC platforms can be straightforward since exportation features is supported by most of the providers to extract generic information of courses learning progression. Currently, this export feature in some case is manually. However, some platforms, such as edX, are currently supporting automatic exportation. ETL pipelines can be particularly developed to transform the custom data mart models to the ICT-FLAG model. With this integration, ICT-FLAG will be able to combine data activity of MOOC platform with ITS interaction.

The LA component's reports (see Fig. 9) include information on how students use the ITS tools and provide instructors with useful information for monitoring and predicting student performance, such as the number/percentage of activities submitted and their

outcomes (total, correct, incorrect); number of attempts needed to solve an activity correctly; amount of time devoted to an activity (from the initial to the last submission).

V. CONCLUSIONS AND FUTURE WORK

This paper has described the current state of learning analytics systems on MOOC platforms. The analysis has showed that the information currently reported is mostly based on aggregated information on assessment and video playback information. Additionally, different initiatives to amend their poor state have been also reviewed.

However, all these approaches are not ready to support external ITS tools. Thus, a design to create a generic learning analytics system to integrate data from MOOC platforms and MOOC-aware ITS has been described. This analytical system could give relevant insights related to the utilization of an ITS in the instructional process of a MOOC course and the relationship with the assessment activities, learning resources and students' performance.

As future work, we will focus on the integration of this analytical system with MOOC platforms and on its test on a MOOC course to check the impact on the course performance analysis.

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